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A Survival–Probability Fusion Prototype for Censor-Aware Multi-Horizon Wildfire Threat Forecasting and Decision-Utility Evaluation

ZhengYang Ren [1,*]

[1] Independent Researcher; d58350215@gmail.com

Abstract

Timely prediction of when a wildfire will reach an evacuation zone supports pre-evacuation readiness and zone triage, yet existing approaches rarely address right censoring, multi-horizon probability consistency, and calibration within a unified decision-support pipeline. We propose a survival–probability fusion framework that unifies a censor-aware survival backbone (XGBoost AFT) with horizon-targeted direct probability heads, post-hoc calibration, and monotone-constrained fusion to produce coherent threat-time probabilities at 12, 24, and 48 h horizons. Using 221 wildfire–zone encounters (69 events, 152 censored; events-per-variable ≈ 2.0) from the WiDS Datathon 2026, nested five-fold cross-validation with inner cross-fitting yielded a pairwise concordance index proxy of 0.940 (Uno's IPCW C-index: 0.941; 95% bootstrap CI: 0.924–0.959), a mean horizon-wise Brier score (IBS proxy) of 0.041, and zero post-fusion monotonicity violations; risk-tercile stratification at 24 h separated observed event rates of 0.0%, 4.5%, and 92.3%. Calibration quality varied across horizons: the 24 h slope approached unity (0.998) while the 12 h horizon exhibited overconfidence (slope = 0.805), and the 48 h horizon was slightly underconfident (slope = 1.075); joint post-fusion temperature scaling improved the 12 h slope to 0.964 without materially degrading other horizons. The full model did not outperform all baselines—a per-horizon XGBoost achieved marginally higher discrimination (0.942) and a Random Survival Forest achieved lower probability error (IBS = 0.032)—indicating that the framework's primary value at this data scale resides in structural guarantees (explicit monotone enforcement, censor-aware formulation, modular extensibility) rather than in aggregate metric superiority. Lean AFT-centred variants (AFT + Cox + Simple: C-index = 0.941, IBS = 0.038) remained competitive with the full prototype, while a repeated five-seed stability analysis confirmed that the Random Survival Forest was the most stable practical comparator under the present sample size (C-index SD = 0.001). Decision curve analysis showed positive net benefit across a wide threshold range at all horizons, supporting the framework's potential decision value. The framework constitutes a methodological prototype; at the current data scale, a simpler AFT-centred architecture with monotone fusion—or a Random Survival Forest—appears more appropriate as an interim practical choice, and although a supplementary proxy incident-grouped validation showed limited optimistic bias (ΔC-index = −0.004), formal incident-level grouped validation with verified identifiers and external spatiotemporal validation remain necessary before operational deployment.

Keywords: wildfire forecasting; evacuation risk; survival analysis; censoring; multi-horizon prediction; calibration; risk stratification; monotone fusion; decision curve analysis

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1. Introduction

Wildfire risk management increasingly requires not merely a binary "will fire arrive" prediction but a probabilistic estimate of *when* a fire will threaten a given evacuation zone. Horizon-specific threat probabilities directly inform pre-evacuation readiness decisions: emergency managers need to rank simultaneously threatened zones by time-criticality, stage suppression and evacuation resources ahead of a fire front, and trigger zone-level alerts with sufficient lead time [1–3]. As the frequency, intensity, and spatial extent of large wildfires have increased over the past two decades [4–10], there is a corresponding demand for decision-support tools that go beyond distance-based heuristics and provide calibrated, multi-horizon risk signals capable of supporting triage under uncertainty [11,12].

Existing approaches to wildfire threat prediction can be broadly grouped into three categories. First, physics-informed estimated-time-of-arrival (ETA) models compute arrival times from distance and fire-spread rate [1,13], but produce point estimates rather than probabilistic outputs and cannot naturally handle right-censored observations, which arise when monitoring ends before fire arrival. Second, single-horizon machine learning classifiers predict a binary outcome at a fixed time window, discarding information from censored cases and failing to ensure probability consistency across multiple windows [14–16]. Third, survival analysis approaches—such as Cox proportional-hazards or accelerated failure time (AFT) models—natively handle censoring and produce survival curves from which horizon-specific cumulative incidence probabilities $P(T \leq h)$ can be derived; these probabilities are inherently monotone non-decreasing by construction [17,18]. However, when *separate* horizon-specific classifiers are trained independently, the resulting probability estimates are not guaranteed to be mutually consistent across horizons, and the survival-derived probabilities may require post-hoc calibration to align with observed event rates at each discrete horizon [19].

Several methodological gaps remain open. First, right-censored observations are common in wildfire encounter datasets when monitoring ends before fire arrival; ignoring them introduces survivorship bias into model training and evaluation, and the modest sample sizes characteristic of labelled wildfire encounter records further require rigorous cross-validation and regularisation strategies to yield generalisable performance estimates [20–22]. Second, operational decision-support requires probability estimates at multiple discrete horizons (e.g., 12, 24, 48 h), and these estimates must be monotonically non-decreasing (i.e., $P(T \leq 12) \leq P(T \leq 24) \leq P(T \leq 48)$) to remain logically coherent across adjacent time windows. Third, discrimination alone (ranking zones by relative threat) is insufficient for threshold-based evacuation triggers; absolute probability calibration is necessary for zone-level decision rules to be actionable [19].

In this study, we present a survival–probability fusion framework designed to address these gaps within an internally validated decision-support prototype. Given the modest sample size available (221 observations, 69 events), the framework is positioned as a methodological contribution demonstrating how censor-aware survival modelling, horizon-targeted probability heads, and monotone enforcement can be integrated within a single pipeline, rather than as a claim of performance superiority over existing methods. The contributions of this work are threefold:

1. A censor-aware multi-horizon prediction framework that unifies an AFT survival backbone with horizon-targeted direct probability heads, post-hoc calibration, and monotone-constrained fusion within a single decision-support pipeline, producing coherent threat-time probabilities at 12, 24, and 48 h—with empirical evidence that explicit monotone enforcement is substantively necessary in stacked multi-horizon setups, correcting logically incoherent probability sequences in approximately 25% of observations at the 24–48 h transition;

2. A leakage-controlled nested cross-validation evaluation protocol with inner cross-fitting that prevents information contamination between meta-feature generation and fusion weight estimation;
3. A transparent evaluation including systematic ablation, baseline comparisons against both survival and non-survival alternatives, decision curve analysis demonstrating positive net benefit across a wide threshold range, and a joint post-fusion recalibration sensitivity analysis—with candid reporting of cases where simpler baselines matched or exceeded the full framework.

2. Materials and Methods

2.1. Study Design and Data Source

This is a retrospective predictive modelling study. The dataset was obtained from the WiDS Datathon 2026 wildfire prediction competition and contains records derived from real wildfire events in the western United States. Each observation represents a single *fire–zone pair*: one wildfire event paired with one evacuation zone, characterised by spatio-temporal and dynamic fire-behaviour variables computed from fire perimeter data observed during the first five hours after ignition. The analysis unit throughout this study is the fire–zone pair; each such pair contributes exactly one observation. The labelled training set comprises 221 fire–zone pairs across 37 variables, with 69 events (fire entering within 5 km of the evacuation-zone centroid) and 152 right-censored observations (monitoring ended before fire arrival or the fire was contained). A held-out test set of 95 observations (35 variables) was available but lacked outcome labels; all model development and evaluation were therefore conducted exclusively using the 221-observation training set under nested cross-validation. All predictor variables were available at or before the end of the five-hour observation window, ensuring no information leakage from the future.

2.2. Outcome Definition and Censoring

The primary outcome is the time from the end of the initial five-hour observation window to the fire perimeter entering within 5 km of the evacuation-zone centroid, measured in hours. This definition follows the WiDS Datathon 2026 task specification: features are derived from fire behaviour observed during the first five hours after ignition, and the prediction target is whether the fire enters a 5 km radius around the zone centroid within 12, 24, or 48 h of the observation window endpoint (Figure 1). Each observation thus represents a single fire–zone pair, with the analysis unit being one wildfire event paired with one evacuation zone. Right censoring occurs when monitoring ends (e.g., fire is contained, observation period lapses) before the fire reaches the 5 km threshold. Notably, censoring arising from fire containment may not be fully non-informative in the statistical sense: containment is an intervention that terminates potential zone impact and is likely correlated with fire behaviour covariates rather than being random. This represents a departure from the standard assumption of independent right censoring, and its potential effect on survival model estimates should be acknowledged as a limitation [20].

Three main prediction horizons—12 h, 24 h, and 48 h—were selected to span the operationally relevant window for evacuation planning: 12 h represents a critical near-term alert threshold, 24 h corresponds to a standard pre-evacuation preparation period, and 48 h supports medium-range resource positioning [3,18]. An additional 72 h horizon was computed but yielded a degenerate positive rate of 100% (all eligible observations experienced the event), rendering it uninformative; results for 72 h are therefore confined to Supplementary Materials (Table S1).

Censor-aware binary targets for each horizon h were defined as follows. An observation is *eligible* at horizon h if either an event occurred at any time or censoring occurred at

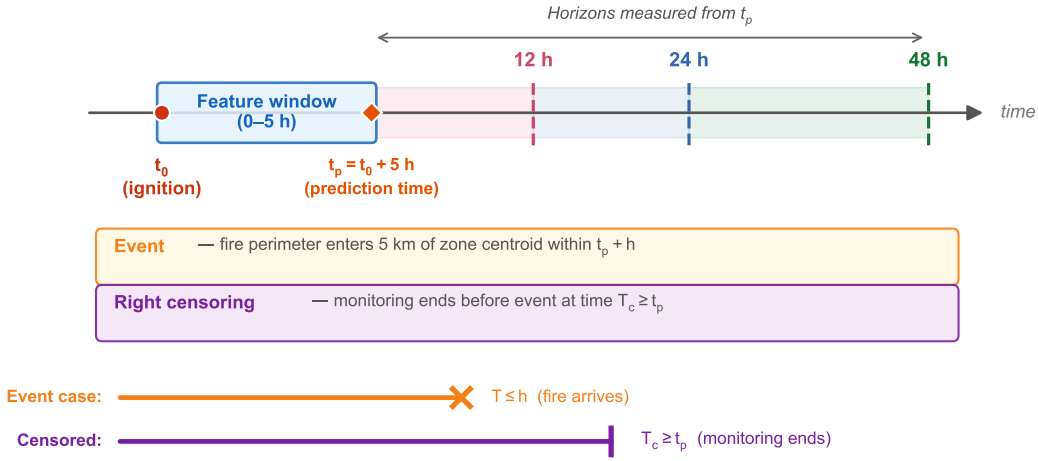

Figure 1. Prediction timeline. Features are extracted from the first five hours after ignition (t_0). The prediction time $t_p = t_0 + 5$ h defines the origin for all horizons. The event is defined as the fire perimeter entering within 5 km of the evacuation-zone centroid. Observations are right-censored if monitoring ends before the event occurs. An "×" denotes an observed event (fire arrival within the horizon); a vertical bar "|" denotes right censoring (monitoring terminated before the event).

a time $\geq h$. Among eligible observations, the target $Y_h = 1$ if the event occurred at time $T \leq h$, and $Y_h = 0$ otherwise. This definition correctly excludes observations that were censored before horizon h and whose true status at h is unknown. This binary formulation does not distinguish between narrowly missed events (e.g., fire arriving at T just beyond h) and clearly non-threatened observations; a continuous-time competing-risks formulation could address this distinction in future work. The eligible sample sizes and positive rates at each horizon are reported in Table 1.

Table 1. Censor-aware target characteristics for each prediction horizon.

Horizon	Eligible n	Positives	Positive Rate (%)
12 h	215	49	22.8
24 h	196	63	32.1
48 h	166	66	39.8
72 h†	69	69	100.0

†Degenerate; confined to Supplementary Materials.

2.3. Predictors and Feature Engineering

Input features were organised into four conceptual groups:

Distance and urgency features.

Raw distance to the closest fire perimeter was transformed into log-distance ($\log(1 + d)$), inverse distance ($1/(1000 + d)$), and a short-range urgency score ($\exp(-d/2500)$). An estimated time of arrival (ETA) was computed as $\text{ETA} = d/\max(v_{\text{closing}}, 50)$, clipped at 240 h.

Speed and alignment features.

Closing speed (signed and absolute), centroid speed, and an alignment measure capturing the angular correspondence between fire spread direction and the vector toward the zone boundary were retained. A directional push interaction was computed as $\max(v_{\text{closing}}, 0) \times |\text{alignment}|$.

Growth and spread dynamics.

Features included area growth rate (ha/h), radial growth rate (m/h), log-transformed growth ($\log(1 + \text{growth})$), and two interaction terms: closing speed $\times$ log-growth and area growth rate $\times$ alignment.

Temporal and regime features.

Cyclical encoding (sine/cosine) was applied to event start hour (period = 24), day of week (period = 7), and month (period = 12). Three binary regime indicators were defined based on distance thresholds: *threat* ($d < 4500$ m), *boundary* ($4500 \leq d \leq 7000$ m), and *far* ($d > 7000$ m). These thresholds were chosen as pragmatic operational cutoffs informed by general fire-management considerations: 4500 m is within the range at which ember transport and spot-fire ignition have been reported to pose structural threat under moderate-to-high wind conditions in coniferous fuel types [1,13], while 7000 m is broadly consistent with outer perimeters used for pre-evacuation staging areas in California wildfire management [23]; however, these values should be regarded as engineering heuristics rather than established domain consensus, and their appropriateness may vary across fuel types and terrain. The ETA computation clipped closing speed at a minimum of 50 m/h (a conservative lower bound for slow-moving ground fires) and capped the estimated arrival time at 240 h (10 days), beyond which the prediction is operationally uninformative [18].

In total, the full feature set comprised 26 engineered features plus 8 retained original dataset columns (after deduplication and exclusion of identifier and outcome variables), yielding 34 features available for modelling (see Supplementary Table S0 for the complete feature dictionary). Missing values were imputed using median substitution for continuous predictors and mode substitution for binary indicators, applied within each outer training fold to prevent leakage; in practice, no variables in the analysis dataset contained missing values (Supplementary Table S11), so the imputation step served as a precautionary safeguard rather than correcting actual missingness. A "thin" feature subset restricted to the 26 engineered features was also evaluated in the ablation study to assess the contribution of original dataset columns.

2.4. Proposed Framework

The proposed framework consists of four modules (Figure 2):

2.4.1. Module A: Gate Prior Estimation

A logistic regression model is trained on the full feature set to predict a binary event indicator. The resulting out-of-fold (OOF) predicted probabilities serve as a *gate prior*—a soft signal capturing the baseline likelihood that a fire will reach the zone—which is subsequently injected as an additional feature into Modules B and C. This provides a soft data-driven summary of overall event likelihood, which may help downstream modules by reducing redundancy among the raw features; its effect is analogous to a stacking-level shrinkage signal, though the degree of regularisation it confers was not independently validated at the present sample size.

2.4.2. Module B: Dual Survival Backbone

Two complementary survival models are trained:

XGBoost AFT ensemble. Three XGBoost [24] AFT models with different distributional assumptions (normal, logistic, extreme value) and hyperparameter settings are fitted. For each observation, the predicted log-survival time from each AFT model is converted to a horizon-specific probability via the corresponding distributional CDF. The ensemble of

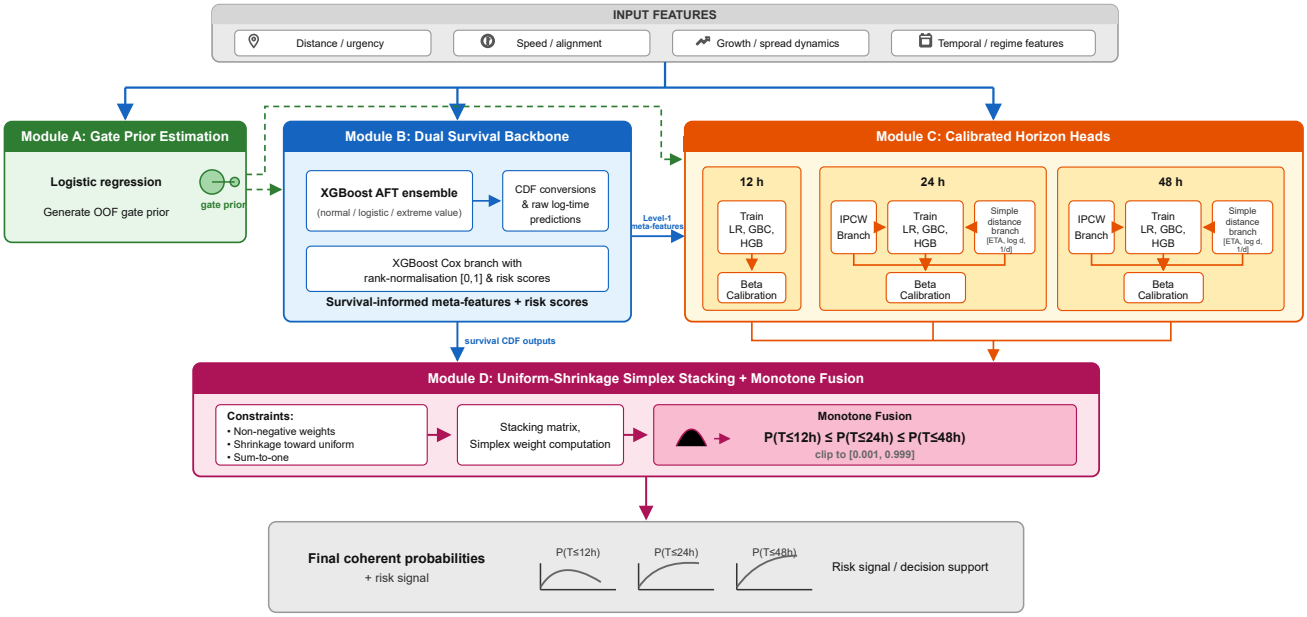

Figure 2. Framework schematic. Module A estimates a gate prior via logistic regression. Module B comprises a dual survival backbone (XGBoost AFT and Cox proportional hazards). Module C trains calibrated direct probability heads (logistic regression, gradient boosting, histogram gradient boosting with beta calibration) at each horizon. Module D performs uniform-shrinkage simplex stacking followed by monotone-constrained fusion to produce final coherent probabilities $P(T \leq h)$ for $h \in \{12, 24, 48\}$.

AFT-derived probabilities, along with the raw log-time predictions, forms a set of survival-informed meta-features.

Cox proportional-hazards ranking branch. Three XGBoost Cox models with varying depth and learning rate configurations are fitted to the same features. The predicted risk scores provide a discrimination-focused signal that complements the AFT branch's emphasis on probability estimation. Risk scores are rank-normalised to the $[0, 1]$ interval before entering the stacking matrix.

Both branches are trained exclusively on inner-fold data with OOF predictions generated via cross-fitting, ensuring that no observation's meta-features depend on a model that saw the same observation during training.

Design note on the Cox branch's role. The Cox branch contributes exclusively to the composite ranking signal used for discrimination analysis and is *not* injected into the horizon-wise probability stacking matrix in Module D. Consequently, adding or removing the Cox branch may change the concordance index (discrimination) but leaves Brier scores, ECE, and calibration slopes unchanged. This separation is by design: the Cox branch serves as a ranking enhancer, not a calibration source.

2.4.3. Module C: Calibrated Horizon Heads

For each horizon $h \in \{12, 24, 48\}$, three direct classifiers—logistic regression (LR), gradient boosting classifier (GBC), and histogram gradient boosting classifier (HGB)—are trained on the censor-aware binary target Y_h. The input feature vector includes all Level 1 meta-features from Modules A and B, alongside the original engineered features. Beta calibration [25] (three-parameter form) is applied post-hoc to each classifier's OOF

probability outputs to improve probability reliability; calibration parameters are estimated on the inner-fold OOF predictions and applied to outer-fold predictions. However, the three-parameter beta calibration may itself be susceptible to overfitting when the effective eligible sample within each inner fold is small (e.g., ∼35 eligible observations at some horizons); a two-parameter or simpler isotonic alternative may prove more stable in future work.

Additionally, at 24 h and 48 h, two supplementary branches are optionally included: (i) an inverse-probability-of-censoring-weighted (IPCW) branch using LightGBM classifiers, which assigns each observation a weight $w_i = \delta_i / \hat{G}(T_i)$, where δ_i is the event indicator and $\hat{G}(T_i)$ is the Kaplan–Meier estimate of the censoring survival function evaluated at the observed time [26,27]; censored observations receive zero weight and thus do not contribute to the IPCW loss, consistent with the standard formulation; and (ii) a simple distance-based branch comprising a logistic regression on the $[\mathrm{ETA}, \log d, 1/d]$ feature triple, providing a physics-informed anchor.

2.4.4. Module D: Uniform-Shrinkage Simplex Stacking and Monotone Fusion

All horizon-specific probability estimates from Module C (and from the survival CDF conversions in Module B) are collected into a stacking matrix [28,29]. Fusion weights are learned by minimising the Brier score [30] on the inner OOF set, subject to non-negativity constraints and regularised toward the uniform distribution (Equation (1)):

$$\mathbf{w}^* = \arg\min_{\mathbf{w} \geq 0} \left[\frac{1}{n} \sum_{i=1}^{n} \left(y_i - \mathbf{w}^\top \mathbf{s}_i \right)^2 + \lambda \left\| \mathbf{w} - \frac{1}{K} \mathbf{1} \right\|_2^2 \right], \tag{1}$$

where $\mathbf{s}_i$ is the vector of stacked probability estimates for observation i, K is the number of base learners, and $\lambda = 1.0$ is a fixed regularisation hyperparameter (not tuned on data) controlling the strength of shrinkage toward the uniform prior. The value $\lambda = 1.0$ was chosen to impose moderate regularisation: it balances the data-driven Brier loss (whose magnitude is $O(10^{-2})$ per observation) against a meaningful pull toward equal weighting, thereby discouraging dominance by any single base learner without suppressing informative weight differentiation entirely. A sensitivity analysis over $\lambda \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$ showed that the primary metrics were stable across this range (see Supplementary Table S2 for details). The optimisation is a convex quadratic program solved via `scipy.optimize.minimize` (L-BFGS-B with bound constraints); convergence was verified across all folds. The weights are subsequently normalised to sum to one.

After stacking, a monotone-constrained fusion step enforces $P(T \leq 12) \leq P(T \leq 24) \leq P(T \leq 48)$ by propagating the maximum cumulatively across horizons and clipping to $[0.001, 0.999]$.

2.5. Validation Design

A nested cross-validation scheme was employed to obtain unbiased performance estimates, following recommendations for prediction model studies [20,31]:

Outer loop (5 folds, stratified by event-time strata): provides the final out-of-fold predictions on which all reported metrics are computed. Stratification was based on discretised event-time quintiles among events and a single stratum for censored observations, ensuring representative event rates in each fold. A caveat is that if multiple observations in the dataset correspond to the same wildfire incident, observations from the same incident may appear in both training and validation folds of the outer loop. The present dataset does not provide a verified incident identifier, precluding a grouped split by fire event. This represents a potential source of optimistic bias: observations from the same fire share

structural similarities (e.g., fire behaviour trajectory, terrain) that could inflate apparent external validity. To partially address this concern, a supplementary proxy incident-grouped cross-validation analysis was conducted using feature-space hierarchical clustering to construct 43 proxy incident groups from the 12 most discriminative covariates; a lean AFT model was then evaluated under both standard stratified and GroupKFold cross-validation to quantify the magnitude of any optimistic bias (see Supplementary Table S8 and Figure S2). A complementary leave-one-month-out temporal blocked cross-validation was also performed as a conservative robustness check (Supplementary Figure S3). These supplementary analyses use a simplified lean AFT model rather than the full multi-module architecture and therefore quantify *relative* bias between validation strategies rather than absolute full-model performance.

Inner loop (5 folds within each outer training set): used for cross-fitting to generate meta-features (Level 1: gate prior, AFT predictions, Cox risk scores; Level 2: direct heads, IPCW, simple branch) and for learning stacking fusion weights. No observation's fusion weight estimation uses meta-features derived from a model that included the same observation in training. A 5×5 configuration was selected as a bias–variance compromise appropriate for the present sample size; smaller fold numbers would increase variance, while larger numbers would leave insufficient inner-training data [20].

Bootstrap confidence intervals (CIs) for the primary metrics were computed using 1000 stratified bootstrap resamples of the outer-fold OOF predictions, with the 2.5th and 97.5th percentiles defining 95% CIs. The number of resamples (1000) was selected to provide stable CI estimates at the observed metric magnitudes [32].

2.6. Baselines and Ablations

Four baselines were evaluated to provide progressively more competitive reference points: (i) *ETA-only*: a logistic regression predicting each horizon target from the ETA feature alone, representing a minimal physics-informed baseline; (ii) *AFT-only*: the three-model AFT ensemble with CDF-derived horizon probabilities and monotone fusion, but without direct heads, Cox, IPCW, or simple branches; (iii) *Per-horizon XGBoost*: a separately trained XGBoost classifier for each horizon using all 34 features on the censor-aware eligible subset, representing a strong single-model machine learning baseline without survival structure; (iv) *Random Survival Forest* (RSF) [33]: 500 trees with log-rank splitting, CDF-derived horizon probabilities, and monotone fusion, representing a competitive nonparametric survival baseline.

The ablation study comprised two complementary designs with distinct inferential purposes. *Layered build-up* evaluated partial configurations of increasing complexity (AFT only, AFT + Direct, AFT + Direct + Cox, AFT + Direct + Gate, Full model) to characterise the contribution of each module group relative to the full model. Note that these configurations represent parallel branches rather than a strictly cumulative sequence: AFT + Direct + Cox and AFT + Direct + Gate each augment the AFT + Direct base with a different module. *Component removal* started from the full model and disabled one module at a time (w/o Gate, w/o Cox, w/o IPCW, w/o Simple Distance, w/o Direct Heads, Thin Features only) to assess whether each module's absence degrades the full-model performance.

For each ablation configuration, a paired delta-metric bootstrap (1000 resamples) was used to test the directional null hypothesis that $\Delta\text{C-index}_{\text{full}-\text{ablated}} = 0$ and $\Delta\text{IBS}_{\text{full}-\text{ablated}} = 0$. One-sided p-values were computed as the proportion of bootstrap resamples in which the full model did not outperform the ablated variant; the directional formulation reflects the pre-specified hypothesis that the full model is at least as good as any reduction [21]. Given the modest sample size, effect sizes and 95% bootstrap confidence intervals are the primary inferential focus; p-values are reported as supplementary

indicators of directional consistency and should be interpreted accordingly rather than as formal hypothesis tests. Two-sided p-values for all ablation comparisons are provided in Supplementary Table S3 for reference.

2.7. Practical Lean Variants and Stability Analysis

Given the low EPV ratio (≈ 2.0), the full multi-module architecture may introduce more estimation variance than the available data can constrain. To assess whether reduced-complexity configurations offer comparable performance, three practical lean variants were evaluated under the same nested cross-validation protocol: (i) *AFT + Cox*: the AFT backbone augmented with the Cox ranking branch only, providing censor-aware survival probabilities with an enhanced discrimination signal; (ii) *AFT + Cox + Simple*: as above, with the addition of the simple distance-based branch at 24 h and 48 h, providing a physics-informed probability anchor; (iii) *AFT + Simple*: the AFT backbone with the simple distance branch but without the Cox ranking branch. Thin-feature variants (restricted to the 26 engineered features) were also evaluated as a supplementary sensitivity analysis but are not included in the main-text narrative because they did not yield more stable performance.

To assess the robustness of model rankings to random-seed variability in cross-validation fold assignment, a multi-seed stability analysis was conducted. All core models—Full Model, AFT + Cox, AFT + Cox + Simple, Per-horizon XGBoost, and RSF—were re-evaluated under five distinct random seeds (42, 123, 314, 2024, 7777), each defining a different stratified fold partition. Mean and standard deviation of C-index and IBS across seeds are reported as stability indicators. This analysis provides a secondary robustness check complementing the supplementary proxy incident-grouped cross-validation (Section 2), which was limited to a lean AFT model due to the absence of verified incident identifiers in the dataset.

2.8. Evaluation Metrics

Discrimination.

Two concordance measures were computed. A pairwise concordance index proxy [34] (Harrell-like pairwise implementation) was computed by comparing all concordant and discordant pairs among eligible observations. Additionally, Uno's IPCW C-statistic [26] was computed to provide a censoring-adjusted discrimination measure; both are reported alongside each other in the primary results, with Uno's C as the preferred measure for censored data.

Calibration and probability error.

Horizon-specific Brier scores [30] were computed on the censor-aware eligible subsets. Expected calibration error (ECE) and maximum calibration error (MCE) were derived from decile-binned reliability diagrams [35]. Calibration slope and intercept were obtained from logistic recalibration of the predicted probabilities against the observed outcomes [19]. A slope of 1.0 and intercept of 0.0 indicate perfect calibration; slopes <1 suggest overconfidence, slopes >1 suggest underconfidence.

Summary error measure.

The mean horizon-wise Brier score (IBS proxy) was computed as the arithmetic mean of the three horizon-specific Brier scores (12, 24, 48 h). This summary measure is used as a proxy for the continuous-time integrated Brier score [27]; it is not a true continuous-time integration, and the distinction is acknowledged accordingly.

Monotonicity.

The number of observations violating $P(T \leq h_1) \leq P(T \leq h_2)$ for each adjacent horizon pair was counted.

Decision utility and decision curve analysis.

To assess the framework's potential value for threshold-based evacuation decisions, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and F1 score were computed at selected probability thresholds (0.1, 0.2, 0.3, 0.5, 0.7) for each horizon. Decision curve analysis (DCA) [43] was conducted by computing net benefit across the full threshold range $[0.01, 0.99]$, comparing the model against "treat-all" (evacuate all zones) and "treat-none" (evacuate no zones) reference strategies. Net benefit is defined as $\mathrm{NB} = \mathrm{TPR} \cdot \pi - \mathrm{FPR} \cdot (1 - \pi) \cdot w$, where π is the event prevalence and $w = p_t/(1 - p_t)$ is the odds at threshold p_t.

2.9. Sensitivity Analysis: Joint Post-Fusion Recalibration

A joint post-fusion recalibration step was applied as a sensitivity analysis to assess whether horizon-specific calibration could be improved after stacking and monotone fusion. Temperature scaling was applied to the 12 h predicted logits (dividing by a temperature parameter T optimised to minimise the 12 h Brier score on OOF predictions via grid search, $T \in [0.5, 3.0]$, step 0.1), followed by re-application of the monotone fusion step to propagate changes to 24 h and 48 h. A legacy 12 h-only temperature-scaling variant (without re-optimisation of longer horizons) was also evaluated for comparison. These steps were *not* used for primary model selection and did not alter the training or evaluation pipeline; they are reported to characterise the calibration trade-off across horizons and inform practical deployment choices.

2.10. Software and Reproducibility

All analyses were conducted in Python 3.10 using scikit-learn 1.3, XGBoost 2.0, LightGBM 4.1, and lifelines 0.28. The primary analysis used a fixed random seed of 42; an additional five-seed stability analysis (seeds 42, 123, 314, 2024, 7777) was conducted to assess robustness of model rankings to fold-assignment variability. Nested cross-validation used $K_{\text{outer}} = K_{\text{inner}} = 5$ stratified folds. The complete analysis code and curated supplementary data files are publicly available at https://github.com/eee-afkk/wids2026-wildfire-survival.

3. Results

3.1. Cohort Characteristics

The study cohort comprised 221 wildfire–zone encounters with a 31.2% event rate (69/221). Table 1 summarises the eligible sample sizes and positive rates at each prediction horizon. The 12 h horizon had the lowest positive rate (22.8%), reflecting the short window available for fire arrival, while the 48 h horizon reached 39.8%. The 72 h horizon was degenerate (100% positives among all 69 eligible observations) and was excluded from the main analysis.

3.2. Primary Model Performance

All primary results are based on the pre-recalibration model evaluated on nested OOF predictions. Table 2 presents the main performance metrics with 95% bootstrap CIs.

The concordance index of 0.940 (Uno's IPCW C-index: 0.941) indicates strong pairwise discrimination, with the model showing favourable relative ranking of threat times across wildfire–zone pairs in over 94% of comparable cases. Brier scores were uniformly low

Table 2. Primary model performance (pre-recalibration, nested OOF evaluation). CI = 95% bootstrap confidence interval (1000 resamples).

Metric	Estimate	95% CI
Pairwise concordance index proxy	0.940	[0.927, 0.955]
Uno's IPCW C-index	0.941	[0.924, 0.959]
Mean Brier (IBS proxy)	0.041	[0.025, 0.058]
Brier @ 12 h	0.060	[0.038, 0.086]
Brier @ 24 h	0.035	[0.017, 0.058]
Brier @ 48 h	0.027	[0.009, 0.050]

across all three horizons, with the 48 h horizon showing the lowest probability error (0.027), followed by 24 h (0.035) and 12 h (0.060).

3.3. Calibration and Reliability

Horizon-specific calibration metrics are presented in Table 3, and reliability diagrams are shown in Figure 3.

Table 3. Calibration summary by prediction horizon (pre-recalibration).

Horizon	n	Brier	ECE	MCE	Cal. Slope	Cal. Intercept
12 h	215	0.060	0.057	0.556	0.805	0.245
24 h	196	0.035	0.014	0.651	0.998	−0.061
48 h	166	0.027	0.031	0.777	1.075	−0.272

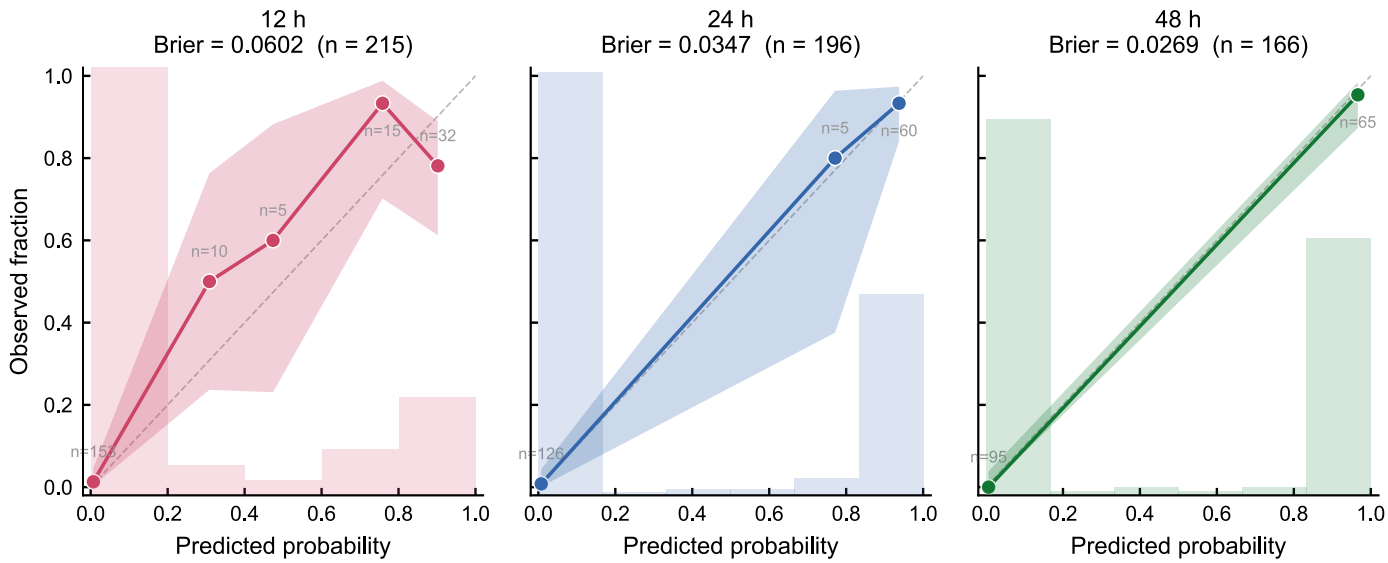

Figure 3. Reliability diagrams for 12, 24, and 48 h horizons (pre-recalibration, nested OOF). Shaded bands represent Wilson confidence intervals. Histograms show predicted probability distributions. The dashed diagonal indicates perfect calibration.

Calibration was best at 24 h (slope = 0.998, essentially at the ideal of 1.0; ECE = 0.014), consistent with the reliability diagram showing near-perfect alignment with the diagonal. At 48 h, calibration was acceptable (slope = 1.075, slightly underconfident; ECE = 0.031). The 12 h horizon was the most challenging, with a calibration slope of 0.805 indicating moderate overconfidence, and an ECE of 0.057. These results are consistent with the well-known difficulty of calibrating models at shorter, more uncertain horizons; the horizon-wise trends in Brier score, ECE, and calibration slope are summarised in Figure 4. Notably, the

maximum calibration error (MCE) increased with longer horizons (0.556 at 12 h to 0.777 at 48 h), despite improving overall calibration; this reflects sparse high-probability bins at longer horizons where large absolute deviations can occur even when average calibration is good. The calibration intercept was positive at 12 h (0.245, indicating underestimation of event probability in the mean) but negative at 24 h (−0.061) and 48 h (−0.272, indicating overestimation); the monotone fusion constraint likely contributes to the 48 h pattern by propagating probabilities forward. From an operational standpoint, the 24 h horizon provides the most reliable absolute probability outputs; the 12 h outputs should be used with caution for threshold-based triggers, and calibration improvement via post-fusion recalibration is assessed as a sensitivity analysis (Section 3.10).

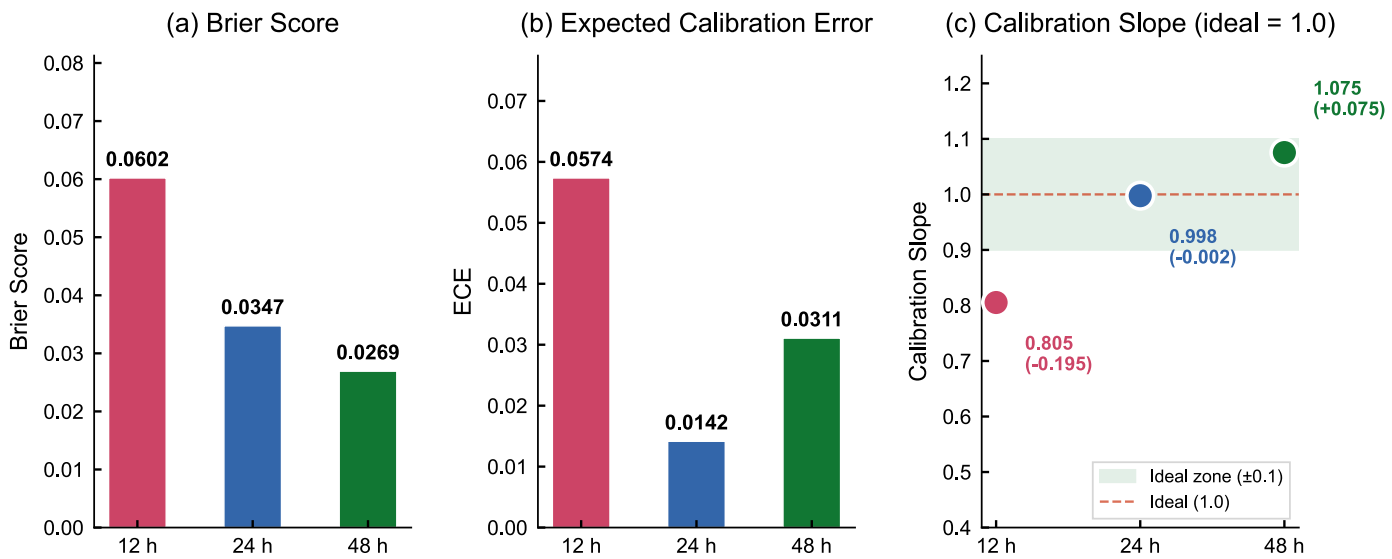

Figure 4. Calibration summary: (a) Brier score, (b) expected calibration error (ECE), and (c) calibration slope by horizon. The dashed line in (c) indicates the ideal slope of 1.0.

3.4. Baseline Comparison

Table 4 and Figure 5 present the baseline comparison.

Table 4. Core model comparison (all models evaluated under the same nested OOF protocol, seed = 42). IBS = mean Brier score across 12/24/48 h horizons. Uno-C = Uno's IPCW C-index. Models are grouped by role: physics-informed baseline, survival-centred lean variants, external comparators, and the full prototype.

Role	Model	C-index	Uno-C	IBS	Brier@12 h	Brier@24 h	Brier@48 h
Baseline	ETA-only (LR)	0.908	0.910	0.045	0.070	0.038	0.028
Lean variants	AFT Only	0.938	0.940	0.039	0.055	0.035	0.027
	AFT + Cox	0.939	0.940	0.039	0.055	0.035	0.027
	AFT + Cox + Simple	0.941	0.942	0.038	0.055	0.035	0.025
External comparators	Per-horizon XGBoost	0.942	0.943	0.040	0.056	0.037	0.026
	Random Survival Forest	0.940	0.941	0.032	0.051	0.028	0.018
Prototype	Full Model	0.940	0.941	0.041	0.060	0.035	0.027

The full model achieved strong discrimination (C-index: 0.940, Uno-C: 0.941) and substantially outperformed the physics-informed ETA-only baseline (ΔC-index = +0.032). However, the full model did not uniformly dominate the stronger alternatives. Table 4 organises the models into three functional tiers to clarify their respective roles.

Among the *lean AFT-centred variants*, AFT + Cox achieved discrimination (C-index: 0.939) close to the full model, confirming the Cox branch's value as a ranking enhancer.

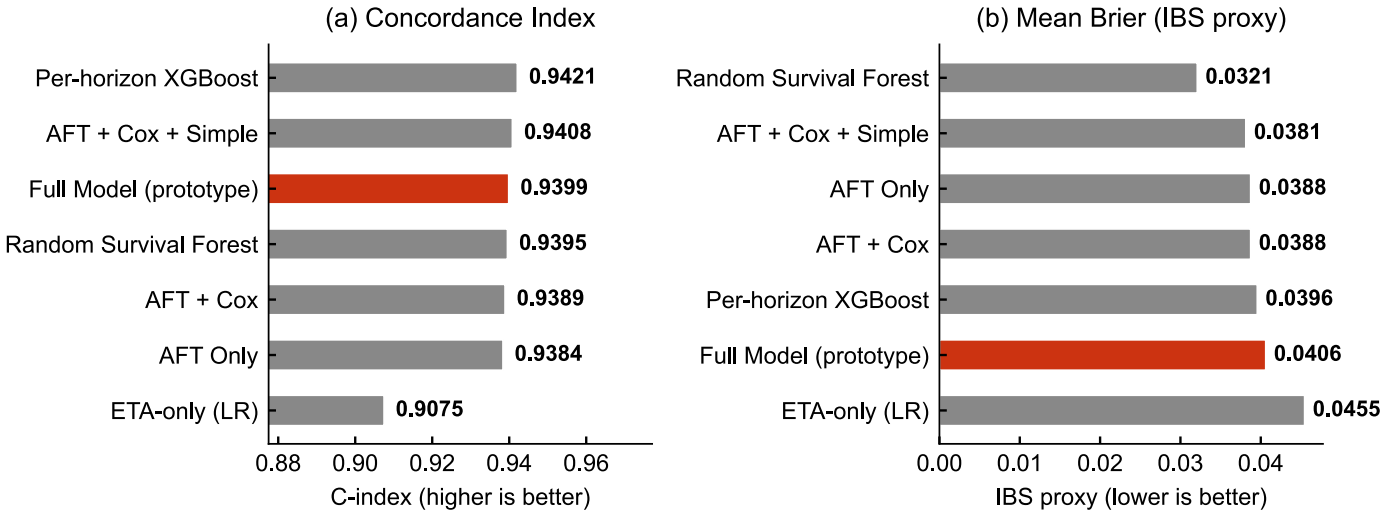

Figure 5. Baseline comparison: (a) pairwise concordance index proxy and (b) integrated Brier score proxy. Higher C-index and lower IBS are better.

AFT + Cox + Simple achieved a lower IBS (0.038) than the full model (0.041), suggesting that the simple distance branch provides a physics-informed anchoring effect on probability error. Notably, AFT Only and AFT + Cox yielded numerically identical Brier scores, ECE, and calibration slopes—as expected from the design in Section 2.4, where the Cox branch enters only the ranking signal and does not alter horizon-specific probability estimates. Likewise, the AFT + Simple variant (described in Section 2.7 but omitted from Table 4 for brevity) yielded numerically identical probability metrics to AFT + Cox + Simple, differing only in C-index.

Among the *external comparators*, the per-horizon XGBoost baseline achieved the highest C-index (0.942) and a comparable IBS (0.040 vs. 0.041), demonstrating that a well-tuned single-model approach without survival structure can match or exceed the full framework in aggregate metrics at this sample size. The Random Survival Forest (C-index: 0.940, IBS: 0.032) achieved the lowest probability error of any model tested, substantially outperforming the full model on IBS despite comparable discrimination.

The full model numerically outperformed the AFT-only baseline (C-index: 0.940 vs. 0.938; IBS: 0.041 vs. 0.039), though the IBS was actually slightly higher for the full model, consistent with the interpretation that additional stacking modules introduce estimation variance at low EPV. From the paired bootstrap analysis (Section 3.5), the ΔC-index between the full model and AFT-only was +0.002 [-0.002, +0.004] ($p = 0.170$), and ΔIBS was -0.002 ($p = 0.805$), both non-significant. These results indicate that the framework's primary value at the present sample size does not reside in aggregate metric superiority over well-tuned alternatives. Rather, the framework offers structural properties—censor-aware formulation, monotone multi-horizon coherence, and modular extensibility—that the per-horizon XGBoost does not provide by construction. The comparison with RSF requires more nuance: RSF is itself a censor-aware survival model whose CDF-derived probabilities are inherently monotone. The framework's distinguishing features relative to RSF lie in its multi-module fusion architecture, horizon-targeted direct probability heads, and modular ablation capability—properties that a monolithic RSF does not offer. Whether these structural properties translate into practical advantages remains to be demonstrated on larger datasets. The RSF's superior IBS may partly reflect the advantage of a single unified survival model on a small dataset, where the additional stacking and fusion stages of the full model introduce estimation variance that the available sample size cannot reliably absorb, consistent with the low EPV ratio (≈ 2.0).

3.5. Ablation Study

Figure 6 presents the ablation forest plot, and Table 5 reports detailed results.

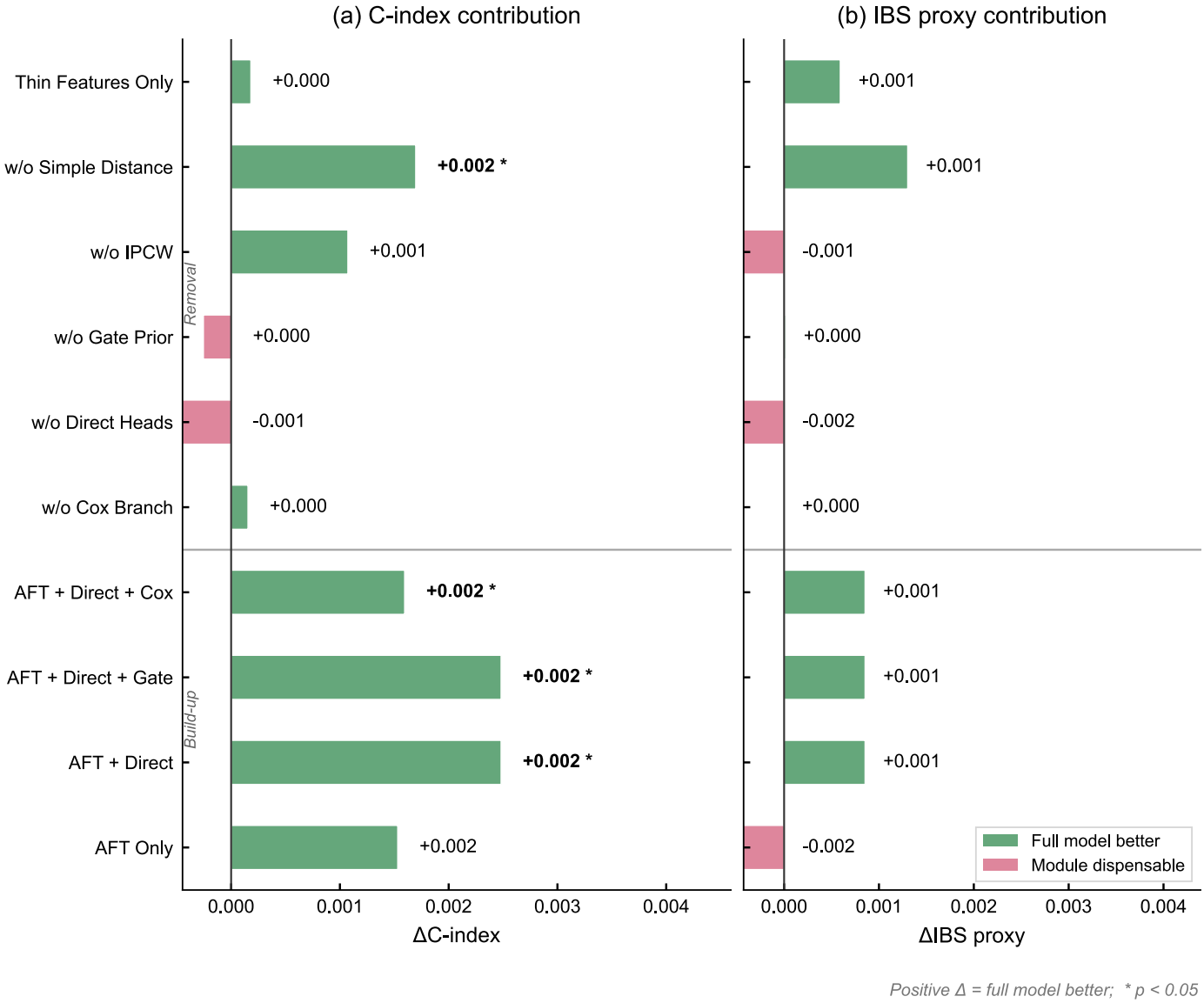

Figure 6. Ablation forest plot showing the contribution of each configuration relative to the full model (ΔC-index and ΔIBS). Positive values indicate that the full model outperformed the ablated variant. Asterisks denote significance at $p < 0.05$ from paired bootstrap.

Among the modular configurations tested in the ablation study, the complete framework showed a numerically favourable discrimination profile, though external baselines such as per-horizon XGBoost and RSF achieved comparable or superior aggregate metrics (see Section 3.4). The ablation results reveal heterogeneous module contributions:

The *Cox branch* had a negligible effect on discrimination in this run: removing it yielded a ΔC-index of +0.000 ($p = 0.892$), while ΔIBS was exactly zero ($p < 0.001$), confirming that the Cox branch affects only the risk ranking signal and does not alter horizon-specific probability estimates.

The *simple distance branch* contributed to both discrimination and probability error: ΔC-index = +0.002 ($p = 0.050$), and ΔIBS = +0.001 ($p = 0.312$), suggesting a physics-informed anchoring contribution.

Table 5. Ablation study: paired delta-metric bootstrap results (1000 resamples). $\Delta > 0$ indicates full model superiority. CI = 95% bootstrap confidence interval. p = one-sided. Significance: * $p < 0.05$.

Configuration	ΔC-index [95% CI]	p	ΔIBS [95% CI]	p
AFT Only	+0.002 [−0.002, +0.004]	0.340	−0.002 [−0.006, +0.003]	0.390
AFT + Direct	+0.002 [+0.000, +0.005]*	0.024	+0.001 [−0.002, +0.004]	0.588
AFT + Direct + Cox	+0.002 [−0.000, +0.003]	0.096	+0.001 [−0.002, +0.004]	0.588
AFT + Direct + Gate	+0.002 [+0.000, +0.005]*	0.024	+0.001 [−0.002, +0.004]	0.588
w/o Gate Prior	−0.000 [−0.001, +0.000]	0.064	+0.000 [−0.000, +0.000]	0.524
w/o Cox Branch	+0.000 [−0.002, +0.002]	0.892	+0.000 [+0.000, +0.000]	<0.001
w/o IPCW	+0.001 [−0.000, +0.002]	0.110	−0.001 [−0.002, +0.001]	0.400
w/o Simple Dist.	+0.002 [+0.000, +0.004]*	0.050	+0.001 [−0.001, +0.004]	0.312
w/o Direct Heads	−0.001 [−0.003, +0.001]	0.422	−0.002 [−0.005, +0.001]	0.126
Thin Features	+0.000 [−0.006, +0.006]	0.964	+0.001 [−0.005, +0.006]	0.834

The *gate prior* and *IPCW* modules showed limited independent incremental gains (ΔC-index and ΔIBS both non-significant when each was individually removed). This does not necessarily indicate these modules are dispensable—their effect may be absorbed by remaining components or may manifest more clearly in larger samples.

The build-up analysis revealed that AFT + Direct exhibited a deficit relative to the full model in C-index ($\Delta = +0.002$, $p = 0.024$), while AFT-only showed a smaller and non-significant deficit ($\Delta = +0.002$, $p = 0.340$). This suggests that adding direct heads without the complementary modules (Cox, gate prior, IPCW, simple distance) may introduce variance; the full model recovers from this by incorporating all modules synergistically. Consistent with this interpretation, removing direct heads from the full model marginally improved IBS ($\Delta = -0.002$, non-significant, $p = 0.126$), reinforcing the view that direct heads introduce estimation variance that is offset by the remaining modules in the full configuration. Among partial configurations, the addition of the Cox branch to AFT + Direct reduced the C-index gap relative to the full model (ΔC-index = 0.002, $p = 0.096$), consistent with the Cox branch's role as a discrimination enhancer.

3.6. Bootstrap Confidence Intervals

Bootstrap distributions for the primary metrics are shown in Figure 7. The C-index distribution (median = 0.940, 95% CI: [0.927, 0.955]) shows tight concentration, indicating stable discrimination. The Uno's IPCW C-index distribution (0.941, [0.924, 0.959]) confirms the censoring-adjusted result is consistent with the proxy. The IBS proxy distribution (0.041, [0.025, 0.058]) and horizon-specific Brier score distributions confirm low probability error with moderate CI width consistent with the sample size. All bootstrap distributions are based on pre-recalibration predictions, ensuring consistency with the primary analysis reported in Table 2.

3.7. Risk Stratification and Monotonicity

After monotone-constrained fusion (Module D), zero violations of $P(T \leq h_1) \leq P(T \leq h_2)$ were observed for both the 12 h → 24 h and 24 h → 48 h transitions across all 221 observations. This result is by construction: Module D enforces monotonicity via cumulative maximum propagation. The more informative diagnostic is the *pre-fusion* violation rate. Prior to monotone fusion, the stacked probabilities exhibited zero violations for the 12 h → 24 h transition and 55/221 (24.9%) for 24 h → 48 h. The mean absolute adjustment imposed by monotone fusion was 0.002 (max: 0.115) for the 24 h → 48 h transition. These non-trivial violation rates indicate that the independently stacked horizon estimates do not naturally respect temporal ordering, and that the monotone fusion step plays a substantive corrective role at the 24 h → 48 h transition rather than serving as a

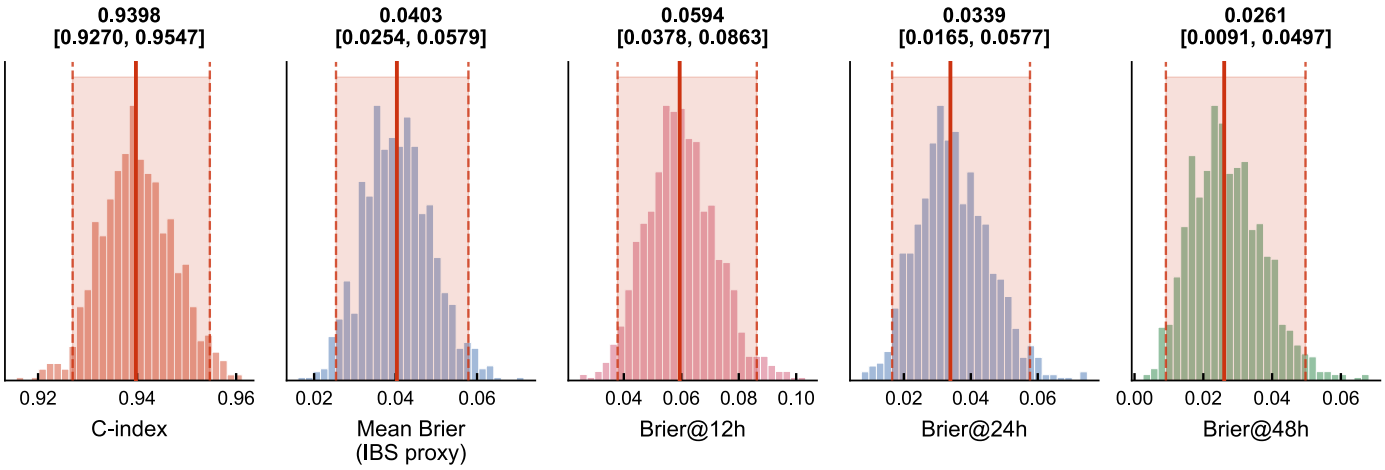

Figure 7. Bootstrap confidence interval distributions (1000 stratified resamples, pre-recalibration) for primary performance metrics. Solid red lines indicate medians; dashed red lines indicate the 2.5th and 97.5th percentiles defining 95% CIs. Shaded bands span the CI region.

minor adjustment. The key finding here is not that the framework produces near-monotone learners, but rather that explicit monotone enforcement is demonstrably necessary in stacked multi-horizon setups: without it, approximately one-quarter of observations at the 24–48 h transition would have received logically incoherent probability sequences. This underscores the importance of the monotonicity enforcement module in the framework design. Scatter plots confirming the post-fusion result are provided in Figure S1 of the Supplementary Materials.

Risk stratification based on 24 h predicted probability terciles produced highly separated observed event rates (Table 6; Figure 8). The low-risk group ($n = 65$) experienced no events within 24 h (0.0%; 95% Wilson CI: 0.0–5.6%), the medium-risk group ($n = 66$) had a 4.5% event rate (CI: 1.6–12.5%), and the high-risk group ($n = 65$) experienced events in 92.3% of cases (CI: 83.2–96.7%). These tercile cutpoints were derived from the OOF predictions and should be regarded as data-dependent; their stability across repeated cross-validation splits has not been formally assessed.

Table 6. Risk stratification by 24 h predicted probability terciles. Event rates and 95% Wilson confidence intervals are computed on the censor-aware eligible subset ($n = 196$).

Risk Tercile	n	Observed Event Rate (%)	95% Wilson CI
Low	65	0.0	[0.0, 5.6]
Medium	66	4.5	[1.6, 12.5]
High	65	92.3	[83.2, 96.7]

3.8. Practical Lean Variants and Model Simplification

Table 4 includes the lean AFT-centred variants alongside the external comparators and the full prototype. The paired bootstrap analysis comparing the full model against each lean variant yielded the following results:

AFT + Cox vs. Full Model: ΔC-index = +0.001 [−0.002, +0.004] ($p = 0.260$); ΔIBS = −0.002 [−0.006, +0.003] ($p = 0.805$); both non-significant. AFT + Cox + Simple vs. Full Model: ΔC-index = −0.001 [−0.004, +0.002] ($p = 0.762$); ΔIBS = −0.002 [−0.007, +0.002] ($p = 0.869$); both non-significant, with the lean variant achieving a numerically lower IBS (0.038 vs. 0.041).

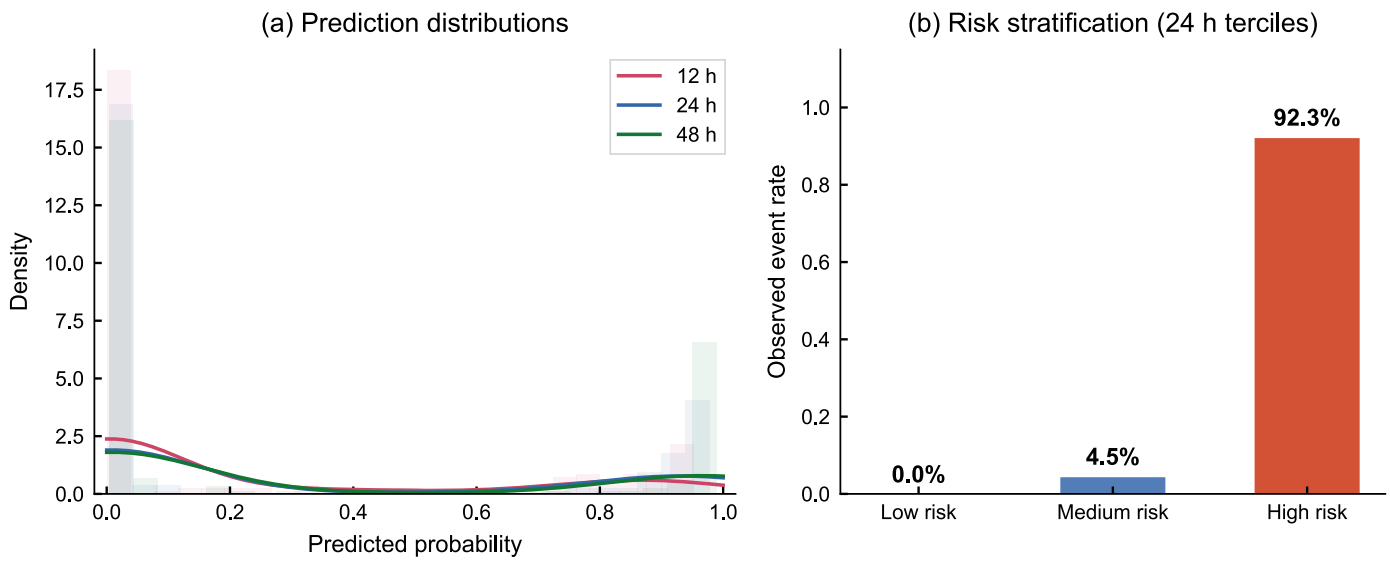

Figure 8. (a) Predicted probability distributions across the three horizons. The bimodal pattern reflects the clear separation between low- and high-threat cases under monotone-constrained fusion, consistent with the strong risk stratification observed in the data. (b) Risk stratification by 24 h predicted probability terciles, with observed event rates of 0.0% (low), 4.5% (medium), and 92.3% (high risk).

These results confirm that the lean AFT-centred variants remain competitive with the full prototype across both discrimination and probability error, and that the additional modules (direct heads, gate prior, IPCW) do not yield statistically significant aggregate gains at the present sample size. The Cox branch's contribution was confined to discrimination: AFT + Cox and AFT Only yielded numerically identical Brier scores (0.039) and calibration metrics, as expected by design, differing only in C-index (0.939 vs. 0.938). Similarly, the simple distance branch improved probability error without affecting discrimination: AFT + Cox + Simple achieved IBS = 0.038 compared with 0.039 for AFT + Cox, while preserving comparable discrimination.

3.9. Multi-Seed Stability Analysis

Table 7 summarises the multi-seed stability results across five random seeds.

Table 7. Multi-seed stability analysis (five seeds: 42, 123, 314, 2024, 7777). Mean and standard deviation of C-index and IBS proxy across seeds.

Model	C-index (mean ± SD)	IBS (mean ± SD)
Random Survival Forest	0.941 ± 0.001	0.032 ± 0.000
Per-horizon XGBoost	0.940 ± 0.001	0.039 ± 0.001
AFT + Cox + Simple	0.939 ± 0.003	0.040 ± 0.001
Full Model (prototype)	0.938 ± 0.003	0.041 ± 0.002
AFT + Cox	0.938 ± 0.003	0.040 ± 0.002

RSF exhibited the smallest variability across seeds on both metrics (C-index SD = 0.001; IBS SD $<$ 0.001), confirming its status as the most stable model under the present sample size. Per-horizon XGBoost was similarly stable (C-index SD = 0.001; IBS SD = 0.001). By contrast, the full model and lean AFT-centred variants exhibited greater seed-to-seed variability (C-index SD = 0.003; IBS SD = 0.001–0.002), consistent with the higher estimation variance expected from multi-module stacking on a small dataset.

The per-seed C-index rankings were informative: RSF ranked first or second in four out of five seeds, whereas the full model's rank fluctuated between first and fourth. The relative ordering between the full model and the lean AFT-centred variants was unstable, with neither consistently dominating the other across seeds. These findings reinforce the conclusion that the performance differences among the survival-based models are within the noise margin at the present sample size, and that RSF represents the most reliable practical comparator.

3.10. Sensitivity Analysis: Joint Post-Fusion Recalibration

The joint post-fusion recalibration sensitivity analysis identified an optimal temperature of $T = 1.50$ for the 12 h horizon. Table 8 and Figure 9 compare pre- and post-recalibration metrics.

Table 8. Pre- vs. post-recalibration comparison. Joint = joint post-fusion temperature scaling; 12h-only = legacy 12 h-only temperature scaling.

	Brier Score			ECE			Cal. Slope	
Horizon	Pre	Joint	12h-only	Pre	Joint	12h-only	Pre	Joint
12 h	0.060	0.058	0.059	0.057	0.048	0.048	0.805	0.964
24 h	0.035	0.035	0.035	0.014	0.010	0.019	0.998	1.014
48 h	0.027	0.027	0.027	0.031	0.031	0.037	1.075	1.052
IBS proxy	0.041	0.040	0.040	—	—	—	—	—

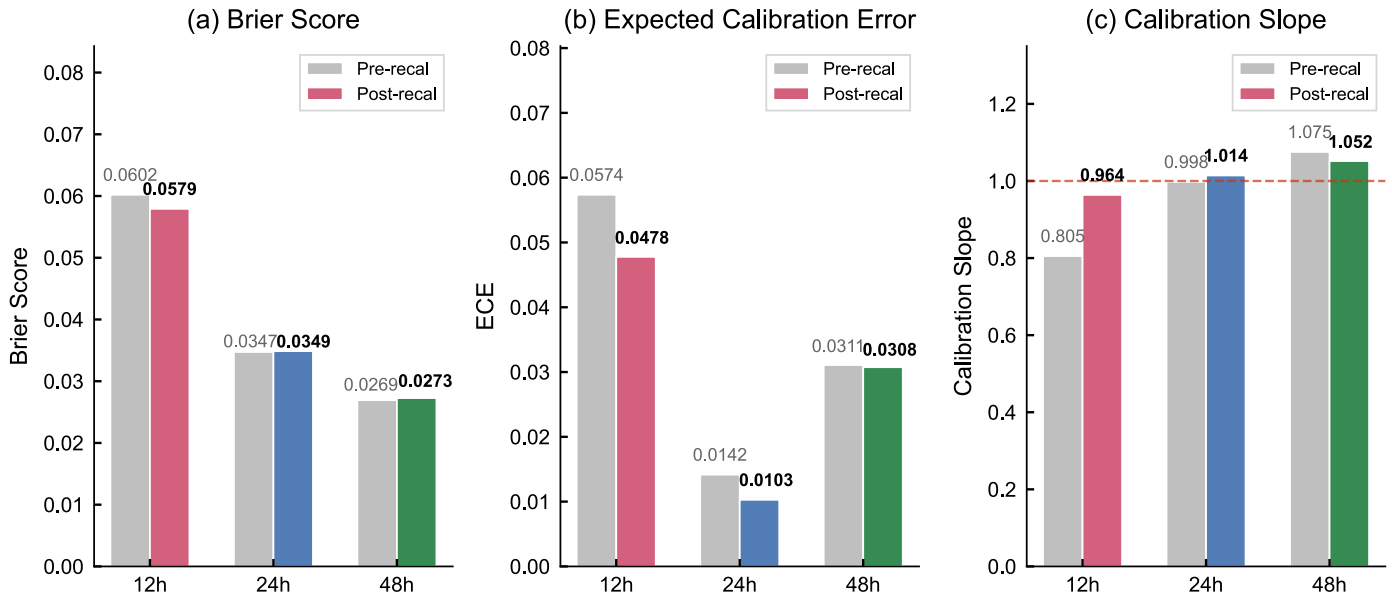

Figure 9. Pre- vs. post-recalibration comparison: (a) Brier score, (b) ECE, and (c) calibration slope. Grey bars represent pre-recalibration values; coloured bars represent post-recalibration values.

Joint post-fusion recalibration improved 12 h short-horizon calibration: the Brier score decreased from 0.060 to 0.058, and the ECE decreased from 0.057 to 0.048. The calibration slope moved from 0.805 (overconfident) to 0.964, close to the ideal value of 1.0. The mean horizon-wise Brier score was slightly reduced from 0.041 to 0.040.

Importantly, the joint recalibration approach preserved 24 h and 48 h calibration: the 24 h ECE improved from 0.014 to 0.010, and the 48 h ECE was essentially unchanged (0.031 to 0.031). By contrast, the legacy 12 h-only temperature scaling produced a more extreme slope correction (1.209, underconfident) and slightly degraded the 24 h ECE (0.019)

and 48 h ECE (0.037). These results demonstrate that joint post-fusion recalibration can meaningfully improve the weakest horizon without materially degrading others, though the improvement is modest and the recalibration is retained as a sensitivity analysis rather than replacing the primary model.

3.11. Decision Curve Analysis and Decision Utility

To evaluate the framework's potential value for threshold-based decision-making, decision curve analysis (DCA) [43] was conducted at all three primary horizons. Table 9 presents decision utility metrics at selected thresholds, and the net benefit across the full threshold range is reported in Supplementary Table S5.

Table 9. Decision utility at selected probability thresholds (pre-recalibration, nested OOF). Sens. = sensitivity; Spec. = specificity; PPV = positive predictive value; NPV = negative predictive value; NB = net benefit.

Horizon	Threshold	Sens.	Spec.	PPV	NPV	F1	NB
12 h	0.1	0.959	0.886	0.712	0.987	0.817	0.209
12 h	0.3	0.939	0.928	0.793	0.981	0.860	0.190
12 h	0.5	0.796	0.946	0.813	0.940	0.804	0.140
24 h	0.1	0.984	0.932	0.873	0.992	0.925	0.311
24 h	0.3	0.984	0.940	0.886	0.992	0.932	0.299
24 h	0.5	0.968	0.955	0.910	0.984	0.938	0.281
48 h	0.1	1.000	0.950	0.930	1.000	0.964	0.394
48 h	0.3	0.985	0.950	0.929	0.990	0.956	0.379
48 h	0.5	0.970	0.960	0.941	0.980	0.955	0.361

The model produced positive net benefit across a wide threshold range at all three horizons, exceeding the "treat-all" and "treat-none" reference strategies. At the 24 h horizon—identified as the best-calibrated—a threshold of 0.3 yielded sensitivity of 0.984, specificity of 0.940, and a net benefit of 0.299, indicating that the model correctly identifies the large majority of threatened zones while maintaining a low false-positive rate. The 48 h horizon showed the highest net benefit (0.394 at threshold 0.1), reflecting the higher base rate and near-perfect sensitivity. The 12 h horizon, despite its weaker calibration, still provided positive net benefit across thresholds up to approximately 0.8, supporting its use for zone-level triage even if absolute probability thresholds require caution. These results indicate that the framework offers measurable decision value beyond what is captured by discrimination and calibration metrics alone.

3.12. Feature Importance

Gain-based feature importance from the AFT backbone (Table 10) revealed that distance-related features dominated the model. Log-distance was the most important predictor (gain = 20.0), followed by minimum distance to the closest perimeter (10.5) and low temporal resolution of perimeter observations (8.9). The top five features accounted for the majority of the total gain, with remaining features contributing incrementally. The dominance of distance-related features is physically intuitive: proximity to the fire front is the strongest predictor of whether a zone will be reached within a given time window. The prominence of temporal resolution and perimeter count features suggests that data quality indicators also carry predictive information, likely serving as proxies for fire complexity or monitoring intensity.

Table 10. Top 10 features by AFT gain-based importance.

Feature	Gain
Log-distance	20.00
Min distance to perimeter	10.48
Low temporal resolution	8.91
Number of perimeters	3.63
Inverse distance	3.42
Time span (first–last)	2.01
Short-range urgency	0.99
Cross-track component	0.89
Month (sine)	0.73
Start month	0.71

4. Discussion

This study presented a survival–probability fusion framework for censor-aware multi-horizon wildfire threat forecasting, evaluated as an internally validated decision-support prototype on a dataset whose modest size (221 observations, 69 events, EPV $\approx$ 2.0) constrains the conclusions that can be drawn about the relative merits of the full multi-module architecture. The framework demonstrated numerically favourable discrimination (concordance index = 0.940, Uno-C = 0.941), low probability error (Brier scores 0.027–0.060 across primary horizons), enforced monotonicity, and strong risk stratification (24 h tercile event rates: 0.0%, 4.5%, 92.3%). Calibration quality varied across horizons: the 24 h horizon was well calibrated (slope = 0.998), the 48 h horizon was slightly underconfident (slope = 1.075), while the 12 h horizon exhibited moderate overconfidence (slope = 0.805); joint post-fusion temperature scaling substantially improved the 12 h slope to 0.964 without materially degrading other horizons. Importantly, the full model did not outperform all alternatives—a per-horizon XGBoost achieved marginally higher discrimination, a Random Survival Forest achieved substantially lower probability error, and lean AFT-centred variants remained competitive—indicating that at this sample size the framework's value resides in its structural design rather than in demonstrated metric superiority. The primary contribution of this work is therefore methodological: the framework provides a principled *architectural template* for combining censor-aware survival modelling with calibrated multi-horizon probability output within a unified decision-support pipeline, and the evaluation transparently characterises both the framework's structural advantages and the conditions under which simpler alternatives may be preferred. Accordingly, the present study should not be read as demonstrating full-model superiority, but rather as establishing a structurally coherent prototype and clarifying when simpler alternatives are preferable. Under the current low-EPV setting, lean AFT-centred variants offer the most defensible censor-aware reductions of the full architecture, whereas RSF provides the strongest immediate practical benchmark in terms of probability error and stability.

Three-tier model roles.

More specifically, the models evaluated in this study fall into three functional tiers. The *full model* serves as a *methodological prototype*: its value lies in demonstrating the integration of censor-aware survival modelling, horizon-targeted probability heads, monotone enforcement, and modular ablation within a single pipeline. The *lean AFT-centred variants* (AFT + Cox, AFT + Cox + Simple) serve as *interim practical alternatives*: they preserve the censor-aware survival formulation and monotone coherence while substantially reducing architectural complexity. AFT + Cox is more oriented toward ranking (C-index: 0.939), while AFT + Cox + Simple additionally benefits from physics-informed probability an-

choring (IBS: 0.038, lower than the full model's 0.041). The *Random Survival Forest* serves as the *strongest practical comparator* at the present sample size: it achieved the lowest IBS (0.032) and the highest stability across repeated random-seed evaluations (C-index SD = 0.001; IBS SD < 0.001). RSF is itself censor-aware and produces naturally monotone survival-derived probabilities; the framework's structural distinction from RSF therefore lies in its multi-module fusion architecture and horizon-targeted calibration pipeline, not in censor-awareness or monotonicity per se.

Framework design and ablation insights.

The integration of survival analysis backbones with calibrated direct probability heads addresses a gap in the wildfire prediction literature. Existing wildfire threat models have largely focused on spatial susceptibility mapping at a single time horizon [14–16,36] or on physics-based fire-spread simulation [1,13,37–39], neither of which produces calibrated multi-horizon probabilities under censoring. Evacuation decision support has been studied through community vulnerability assessments and trigger-point analyses [23,40,41], and time-to-event perspectives have been discussed in the broader wildfire risk literature [2,18], but zone-level threat prediction with calibrated multi-horizon probabilistic output under censoring has not been previously addressed. The present framework bridges these gaps: the AFT backbone provides a principled time-to-event prior that is naturally censor-aware, while the direct heads contribute horizon-specific probability refinement.

A key empirical finding is that explicit monotone enforcement is demonstrably necessary in stacked multi-horizon setups: prior to fusion, approximately 25% of observations at the 24–48 h transition exhibited logically incoherent probability sequences, confirming that independently stacked horizon estimates do not naturally respect temporal ordering. The ablation results further indicate that the direct heads require the complementary modules to function effectively: the AFT + Direct configuration showed a significant C-index deficit relative to the full model (ΔC-index = +0.002, $p = 0.024$), while AFT-only showed a comparable but non-significant deficit (ΔC-index = +0.002, $p = 0.340$), suggesting that direct heads alone may introduce variance that the remaining modules subsequently compensate for. This synergistic pattern supports a modular design in which individual component contributions are not strictly additive.

Module prioritisation under low EPV.

The combined evidence from the ablation study, lean-variant comparison, and multi-seed stability analysis permits a more informed—though still provisional—assessment of module contributions at the present data scale. The *Cox branch* showed a negligible independent effect on discrimination in the primary analysis: removing it yielded a ΔC-index of +0.000 ($p = 0.892$), while ΔIBS was exactly zero, confirming that the Cox branch affects only the ranking signal by design; however, the lean-variant comparison showed that adding Cox to AFT improved C-index from 0.938 to 0.939. The *simple distance branch* contributed to probability error reduction: AFT + Cox + Simple achieved IBS = 0.038 compared with 0.039 for AFT + Cox, and the ablation w/o Simple Distance reached borderline significance (ΔC-index = +0.002, $p = 0.050$). The *gate prior* and *IPCW* modules showed limited independent incremental gains at the current sample size (ΔC-index and ΔIBS both non-significant when individually removed); their effects may be absorbed by remaining components or may manifest more clearly in larger samples. The *direct heads* did not demonstrate stable incremental benefit: removing them from the full model marginally improved IBS ($\Delta = -0.002$, $p = 0.126$), reinforcing the interpretation that direct heads introduce estimation variance that is offset by the remaining modules in the full configuration but that may not be justified at the current EPV. These results suggest that

the most defensible reduced architecture at EPV ≈ 2.0 centres on the AFT backbone with optional Cox (for discrimination) and simple distance (for probability anchoring), while the remaining modules should be regarded as extensions whose incremental value requires confirmation on larger datasets.

Calibration interpretation.

The metrics in Table 3 refer to the final fused probabilities after stacking and monotone fusion, not to the individual base heads. The 12 h horizon is inherently the most difficult to calibrate, as it covers a short and highly stochastic window; the pre-recalibration slope of 0.805 indicates moderate overconfidence, meaning that threshold-based evacuation triggers at 12 h would benefit from recalibration. The 24 h slope of 0.998 is essentially at the ideal and suggests that 24 h probability outputs are directly usable for threshold-based decisions. The 48 h slope of 1.075 indicates slight underconfidence; probabilities are marginally conservative. A structural factor contributes to the 12 h pattern: the beta calibration in Module C targets each base head independently, but the subsequent stacking and monotone fusion steps alter the probability distribution. Joint post-fusion temperature scaling successfully corrected the 12 h slope to 0.964 while preserving the 24 h and 48 h calibration (Table 8), demonstrating that targeted post-fusion recalibration can address the weakest horizon without the cascading degradation seen in the legacy 12 h-only approach. This confirms that multi-horizon frameworks require joint recalibration strategies [35].

Horizon-level deployment readiness.

Based on internal validation only, the 24 h horizon (slope = 0.998) is the most promising for eventual operational use, with near-ideal calibration; the 48 h horizon (slope = 1.075) is slightly underconfident but usable with conservative margins; and the 12 h horizon (slope = 0.805) exhibits moderate overconfidence that restricts its use to relative ranking or requires post-fusion recalibration before threshold-based triggers. Improving 12 h calibration without degrading longer horizons requires addressing the cascading constraint imposed by monotone fusion; the joint post-fusion temperature scaling evaluated in this study (Section 3.10) successfully improved the 12 h slope to 0.964, and further directions include bidirectional monotone adjustment and replacing cumulative-maximum enforcement with a soft monotone penalty during stacking optimisation.

Practical model selection at this data scale.

At the present data scale ($n = 221$, 69 events; EPV ≈ 2.0), the full framework's architectural advantages—censor-aware formulation, explicit monotone enforcement, modular ablation transparency—did not translate into statistically significant aggregate gains over leaner alternatives (ΔC-index vs. AFT-only: $+0.002$, $p = 0.340$; ΔIBS: -0.002, $p = 0.390$). The lean AFT-centred variants achieved comparable discrimination and probability error with far simpler architectures, while RSF combined the lowest IBS (0.032) with the highest seed-to-seed stability. At low EPV, the most defensible interim choices are therefore lean AFT-centred variants when modularity is desired, or RSF when predictive robustness is prioritised; the additional modules in the full architecture should be regarded as research-oriented extensions whose value requires confirmation on larger datasets.

Supplementary proxy grouped and temporal validation.

To partially address the absence of verified incident identifiers, a supplementary proxy incident-grouped cross-validation was conducted using feature-space hierarchical clustering to construct 43 proxy groups (Supplementary Figure S2, Table S8). Under a lean AFT model, the grouped CV C-index (0.928 ± 0.029) was only modestly lower than standard stratified CV (0.932 ± 0.011; $\Delta = -0.004$), suggesting that within-group information leakage

contributes limited optimistic bias at the present data scale. A complementary leave-one-month-out temporal blocked CV (Supplementary Figure S3) yielded an N-weighted mean C-index of 0.935 across the six months with sufficient sample size ($n \geq 10$); the four peak fire-season months (Jun–Sep, $n = 33$–70 each) ranged from $C = 0.887$ to 0.943, while the two shoulder-season months (Mar, May; $n = 14$ each) produced less stable estimates including a likely artefactual $C = 1.000$ for March. These results are supportive of temporal robustness during peak fire season but should be interpreted cautiously given the small per-month sample sizes. A supplementary worst-case/best-case event-rate bounding analysis for informative censoring (Supplementary Figure S4) showed sensitivity ranges of 0.027 (12 h), 0.113 (24 h), and 0.249 (48 h), confirming that the non-informative censoring assumption is tenable at short horizons but warrants caution at 48 h where 55 observations were censored before the horizon. These supplementary analyses use a simplified lean AFT model rather than the full architecture, and the proxy grouping is based on feature similarity rather than ground-truth incident labels; the results therefore represent a lower bound on the stringency of the robustness check. Formal incident-level grouped validation with verified identifiers remains an important priority for future work.

Illustrative operational scenario.

For example, a wildfire–zone pair at 3200 m with a closing speed of 400 m/h and growth rate of 50 ha/h would yield fused probabilities of approximately $P(T \leq 12) = 0.72$, $P(T \leq 24) = 0.91$, $P(T \leq 48) = 0.95$, placing it in the high-risk tercile and supporting immediate pre-evacuation preparation based on the better-calibrated longer horizons; the 12 h probability would serve only for relative prioritisation among simultaneously threatened zones.

Practical implications for wildfire management.

The risk stratification capability may represent the most actionable contribution of this work, as it does not depend on absolute probability calibration. The 24 h tercile separation (0.0% vs. 92.3% observed event rates) supports a straightforward triage protocol: zones in the high-risk tercile can be prioritised for immediate pre-evacuation preparation and resource staging, medium-risk zones can be placed on standby with elevated monitoring frequency, and low-risk zones can remain at routine surveillance intensity. This tercile-based approach is more robust than raw probability thresholds because it relies on the model's ranking ability (C-index = 0.941) rather than on its absolute calibration, which varies across horizons as documented above. Decision curve analysis (Section 3.11) confirmed positive net benefit across a wide threshold range at all three horizons, and the decision utility evaluation (Table 9) showed that at operationally relevant thresholds (e.g., 0.3 at 24 h) the model achieved sensitivity > 0.98 and specificity > 0.93, supporting its potential value for zone-level triage. Further evaluation against agency-specific decision thresholds in operational settings represents an important next step before practical deployment. The framework is positioned as a decision-support tool; final evacuation orders would rest with human responders integrating multiple information streams, and deployment would require integration with near-real-time fire-behaviour feeds [11].

Strengths and limitations.

The strengths of this study include the censor-aware problem formulation, the nested cross-validation protocol with inner cross-fitting that prevents information leakage, and the enforcement of multi-horizon monotonicity. The study further benefits from transparent reporting of metric proxy definitions, comprehensive baseline and ablation analyses, evaluation of practical lean variants that identify defensible reduced architectures, and a

five-seed stability analysis providing a secondary robustness check. The candid acknowledgement that simpler alternatives matched or exceeded the full framework at this sample size is itself informative for practitioners deciding which modelling approach to adopt. Nevertheless, the baseline comparisons could be further strengthened: the per-horizon XGBoost and RSF baselines included here are informative but not exhaustive. Future evaluations should additionally include deep survival models (e.g., DeepSurv [42]) and discrete-time hazard models, which would provide a more demanding comparison against which the framework's structural advantages can be assessed.

Several specific limitations must be acknowledged. First, and most consequentially, the sample size ($n = 221$, 69 events) is insufficient to reliably support the full modelling complexity (34 features, multiple stacked modules). With 69 events and 34 candidate features, the events-per-variable (EPV) ratio is approximately 2.0, well below the commonly recommended threshold of EPV $\geq$ 10 [22]. This low EPV is the most parsimonious explanation for the central empirical finding that the full model did not significantly outperform the AFT-only baseline or simpler competitors: the additional modules likely introduce more estimation variance than the available data can constrain. The ablation results reinforce this interpretation: several modules (gate prior, IPCW) showed negligible independent contributions, and simpler baselines (per-horizon XGBoost, RSF) matched or exceeded the full model on aggregate metrics. The full multi-module framework should therefore be understood as an architectural demonstration whose practical benefits remain to be established on datasets with substantially higher EPV. The low EPV also raises concerns about the stability of the three-parameter beta calibration applied within inner folds, where effective eligible sample sizes may be as low as $\sim$35 observations at some horizons. This limits the precision of bootstrap CIs, the statistical power to detect small ablation effects, and the generalisability of all performance estimates. Second, the outer cross-validation folds were stratified by event-time strata but not grouped by wildfire incident identifier. If multiple zone observations correspond to the same underlying fire event, structurally similar observations may appear in both training and validation splits of the outer loop, potentially introducing optimistic bias into the reported performance metrics. Fold-level performance variability (reported in Supplementary Table S4) should be considered when interpreting aggregate metrics. A supplementary proxy incident-grouped cross-validation—using feature-space hierarchical clustering to construct 43 proxy groups from 12 covariates—showed a modest C-index decrease from 0.932 (standard stratified CV) to 0.928 (grouped CV; $\Delta = -0.004$) under a lean AFT model, suggesting that within-cluster correlation introduces limited optimistic bias at the present data scale (Supplementary Figure S2, Table S8). However, this analysis used proxy groups derived from feature similarity rather than verified incident identifiers, and the grouping quality is inherently approximate; formal incident-level grouped validation remains necessary when verified identifiers become available. Third, no external spatiotemporal validation was conducted; all results are from internal validation and may not generalise to wildfires in different geographic or climatic regions. Fourth, the censoring mechanism may not be fully non-informative: fire containment, one of the two main censoring causes, is an intervention correlated with fire behaviour and therefore may not satisfy the standard independent right-censoring assumption. A supplementary worst-case/best-case event-rate bounding analysis quantified the sensitivity of horizon-specific event rates to this assumption: the sensitivity range was negligible at 12 h (0.027; 6 observations censored before horizon), moderate at 24 h (0.113; 25 censored), and substantial at 48 h (0.249; 55 censored), indicating that the non-informative censoring assumption is defensible at short horizons but materially affects the 48 h event-rate estimate, where the true positive rate could plausibly range from 29.9% to 54.8% depending on the censoring mechanism

(Supplementary Figure S4, Table S9). Consequently, 48 h probability outputs should be interpreted with greater caution than shorter-horizon estimates, and future work should explore censoring-robust methods such as sensitivity-weighted survival models or explicit competing-risk formulations that treat containment as a distinct event type. Fifth, the 72 h horizon was degenerate and could not be meaningfully evaluated. Sixth, the primary concordance index is a pairwise proxy [34]; although Uno's IPCW C-statistic [26] is also reported and yields consistent results, both estimators have limitations under the present censoring distribution. Similarly, the IBS proxy is the arithmetic mean of discrete-horizon Brier scores, not a continuous-time integration [27]. These approximations were adopted for computational tractability under the present sample size and should be interpreted accordingly. Seventh, the post-hoc temperature-scaling recalibration is a sensitivity analysis only, not a core model component, and it introduces a calibration trade-off across horizons. Finally, this study represents an internally validated decision-support prototype; direct deployment for autonomous evacuation decision-making would require incident-level validation, external validation across diverse fire regimes, real-time data integration, and regulatory review.

5. Conclusions

A survival–probability fusion framework was developed and internally validated for multi-horizon wildfire threat forecasting under right censoring. The framework produced coherent, monotonic probability estimates at 12, 24, and 48 h horizons with strong discrimination (concordance index 0.940, Uno-C 0.941), low probability error (Brier scores 0.027–0.060), and effective risk stratification (24 h tercile event rates: 0.0%, 4.5%, 92.3%). Calibration was horizon-dependent: the 24 h horizon was well calibrated (slope = 0.998), the 48 h horizon was slightly underconfident (slope = 1.075), but the 12 h horizon exhibited moderate overconfidence (slope = 0.805); joint post-fusion temperature scaling improved the 12 h slope to 0.964 without materially degrading other horizons.

The primary contribution is methodological rather than performance-oriented. The framework establishes a structured architectural template for censor-aware multi-horizon probability estimation with explicit monotone enforcement—an enforcement the ablation study confirmed to be substantively necessary, correcting logically incoherent probability sequences in approximately 25% of observations. However, the full model did not outperform all baselines in aggregate: a Random Survival Forest achieved substantially lower probability error and the highest seed-to-seed stability, while lean AFT-centred variants matched the prototype's discrimination with far simpler architectures. At the present data scale ($n = 221$, 69 events, EPV ≈ 2.0), a simpler AFT-centred architecture with monotone fusion—or a Random Survival Forest—appears more appropriate as an interim practical choice; the full multi-module architecture constitutes a research prototype whose incremental benefits should be confirmed on larger datasets. Decision curve analysis confirmed positive net benefit across a wide threshold range at all three horizons, supporting the framework's potential value for zone-level triage even at the current sample size.

Future work should prioritise formal incident-level grouped cross-validation with verified identifiers—a supplementary proxy analysis using feature-space clustering showed limited optimistic bias (ΔC-index = -0.004), but the grouping is approximate and should not substitute for ground-truth incident labels—as well as external spatiotemporal validation across diverse fire regimes, joint post-fusion calibration strategies that simultaneously address all horizons under the monotone constraint, and comparison with discrete-time hazard formulations and deep survival models (e.g., DeepSurv [42]). A supplementary informative censoring sensitivity analysis showed event-rate sensitivity ranges of up to 24.9 percentage points at the 48 h horizon, motivating future work on censoring-robust

estimation methods for wildfire applications. Prospective evaluation in operational early warning settings, integrated with real-time fire-spread data feeds, is the necessary next step toward deployment.

Supplementary Materials: The following supporting information is available in the public repository at https://github.com/eee-afkk/wids2026-wildfire-survival: Table S0: Complete feature dictionary (name, type, transformation, source). Table S1: 72 h horizon performance results. Table S2: Full hyperparameter configurations for all models, including stacking regularisation sensitivity analysis ($\lambda \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$). Table S3: Two-sided p-values for all ablation paired bootstrap comparisons. Table S4: Fold-level performance metrics (C-index and Brier scores per outer fold). Table S5: Decision curve analysis net benefit across the full threshold range at 12, 24, and 48 h. Table S6: Complete feature importance ranking (all features). Table S7: Practical lean variant comparison with thin-feature configurations. Table S8: Proxy incident-grouped cross-validation comparison (standard stratified CV vs. feature-space clustering GroupKFold; lean AFT model). Table S9: Informative censoring sensitivity analysis—worst-case/best-case event-rate bounds and Aalen–Johansen competing-risk cumulative incidence by horizon. Table S10: Leave-one-month-out temporal blocked cross-validation results (lean AFT model). Table S11: TRIPOD-style data quality summary (missing rates, variable distributions). Figure S1: Monotonicity verification scatter plots. Figure S2: Proxy incident-grouped CV comparison—boxplots of C-index and mean Brier score under standard stratified vs. grouped cross-validation (43 feature-space proxy groups). Figure S3: Leave-one-month-out temporal blocked CV—per-month C-index and mean Brier score (months with $n < 10$ excluded); the figure reports unweighted monthly means ($\bar{C} = 0.937$), whereas the main text reports the N-weighted mean ($C = 0.935$) to account for differing per-month sample sizes. Figure S4: Informative censoring sensitivity—worst/best-case event-rate bounds and Kaplan–Meier vs. Aalen–Johansen cumulative incidence by horizon. Figure S5: Kaplan–Meier survival function and event-time vs. censoring-time distributions. Figure S6: Eligible sample size and event rate per prediction horizon. Figure S7: NIFC WFIGS geographic context—western US wildfire incidents used for supplementary proxy incident grouping (note: the 2026 bar in the annual counts panel may represent a partial year in the downloaded open-data snapshot and should not be interpreted as an anomalous decline).

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The following abbreviations are used in this manuscript: AFT = Accelerated failure time; CDF = Cumulative distribution function; CI = Confidence interval; DCA = Decision curve analysis; ECE = Expected calibration error; ETA = Estimated time of arrival; GBC = Gradient boosting classifier; HGB = Histogram gradient boosting classifier; IBS = Integrated Brier score; IPCW = Inverse probability of censoring weighting; LR = Logistic regression; MCE = Maximum calibration error; NB = Net benefit;

NPV = Negative predictive value; OOF = Out-of-fold; PPV = Positive predictive value; RSF = Random Survival Forest.

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